Where Does the Energy Go?

Aim: To find out where the heat energy from hot water goes to as the water cools down.

Materials: Ice-cream container with cold water in it; 250 mL Beaker; 2 thermometers, boiling water from a kettle (be careful, hot water is dangerous), stop watch; graph paper.

Method:

1. Half fill your 250 mL beaker with cold water.
2. Place the beaker in your ice cream container and add cold water to the ice cream container until it covers the cold water in your beaker.
3. Take the beaker out of the water and tip out the cold water.
4. Take the temperature of the water in the ice cream container and record in your table,
5. Get your 250 mL beaker and have it half filled with hot water.
6. Quickly measure the temperature of this water and record it in your table.
7. Put the beaker into the ice cream container.
8. Start your stop watch and record the temperature of the water in the beaker and the temperature of the water in the ice cream container every 30 seconds for 13 minutes.

Results:

Time (min)	Temperature of Beaker water (˚C)	Temperature of ice-cream container water (˚C)	Time (min)	Temperature of Beaker water (˚C)	Temperature of ice-cream container water (˚C)
0			7		
0.5			7.5		
1			8		
1.5			8.5		
2			9		
2.5			9.5		
3			10		
3.5			10.5		
4			11		
4.5			11.5		
5			12		
5.5			12.5		
6			13		
6.5			13.5		

Questions:

1. What happened in the experiment: explain what happened to the heat energy in the hot water.

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2. Use your graph paper to make a graph of the results. You will need to plot the time on the x-axis (horizontally) and Temperature on the Y-axis. Use a different colour to plot the water in the beaker and another to plot the temperature of the water in the ice-cream container.

Do This question on your graph paper.

3. On your graph, add a dotted line to show what you predict will continue to happen if time was allowed to go longer.

Do This question on your graph paper.

4. From your results, write a sentence that explains what Conservation of Energy means.
